Topsoil Degradation

Environmental Economics (HSS504)
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The Scale of the Problem

~ 30%

of the world's soil is moderately to highly degraded (UNDRR, 2024)

24B

tonnes of fertile topsoil lost globally each year (United Nations, 2019)

95%

of food depends on healthy soil (FOA, 2024)

The UN Food and Agriculture Organization (FAO, 2014) estimates that if current rates of soil degradation continue, the world could run out of topsoil within 60 years.

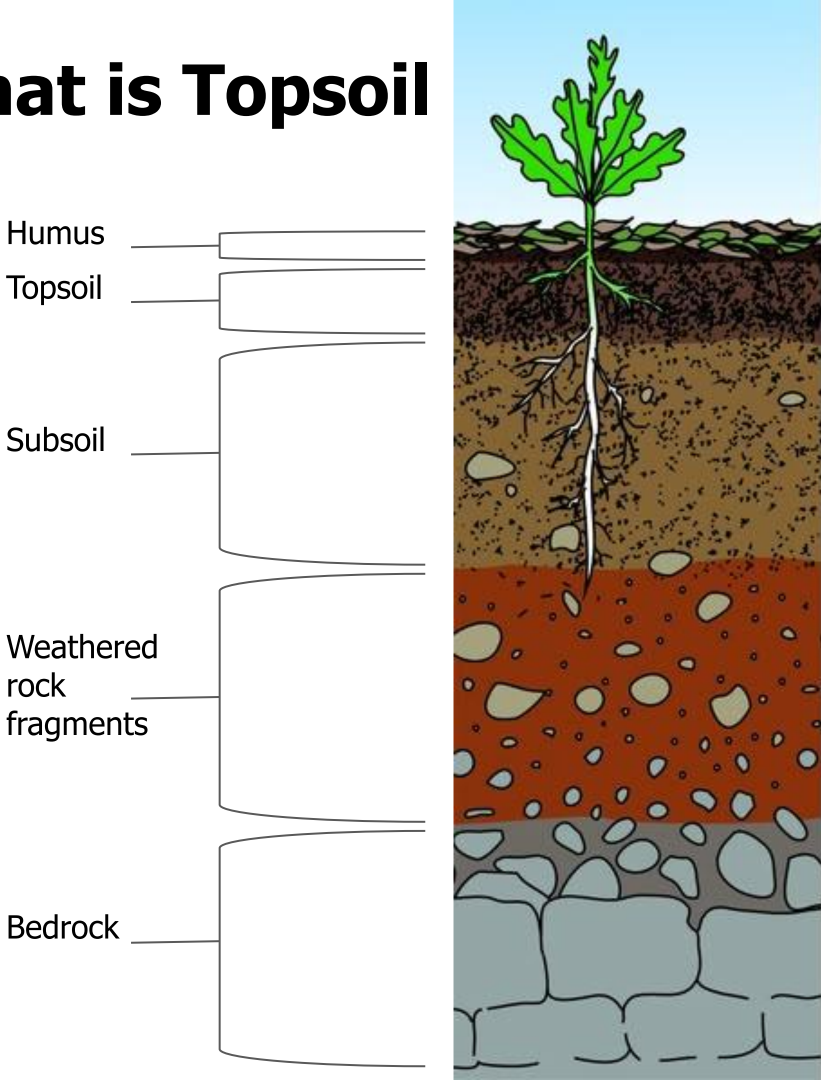
What is Topsoil

Formation rate: ~1 cm topsoil per 200-400 year under natural conditions. (JIC, 2019/2020)

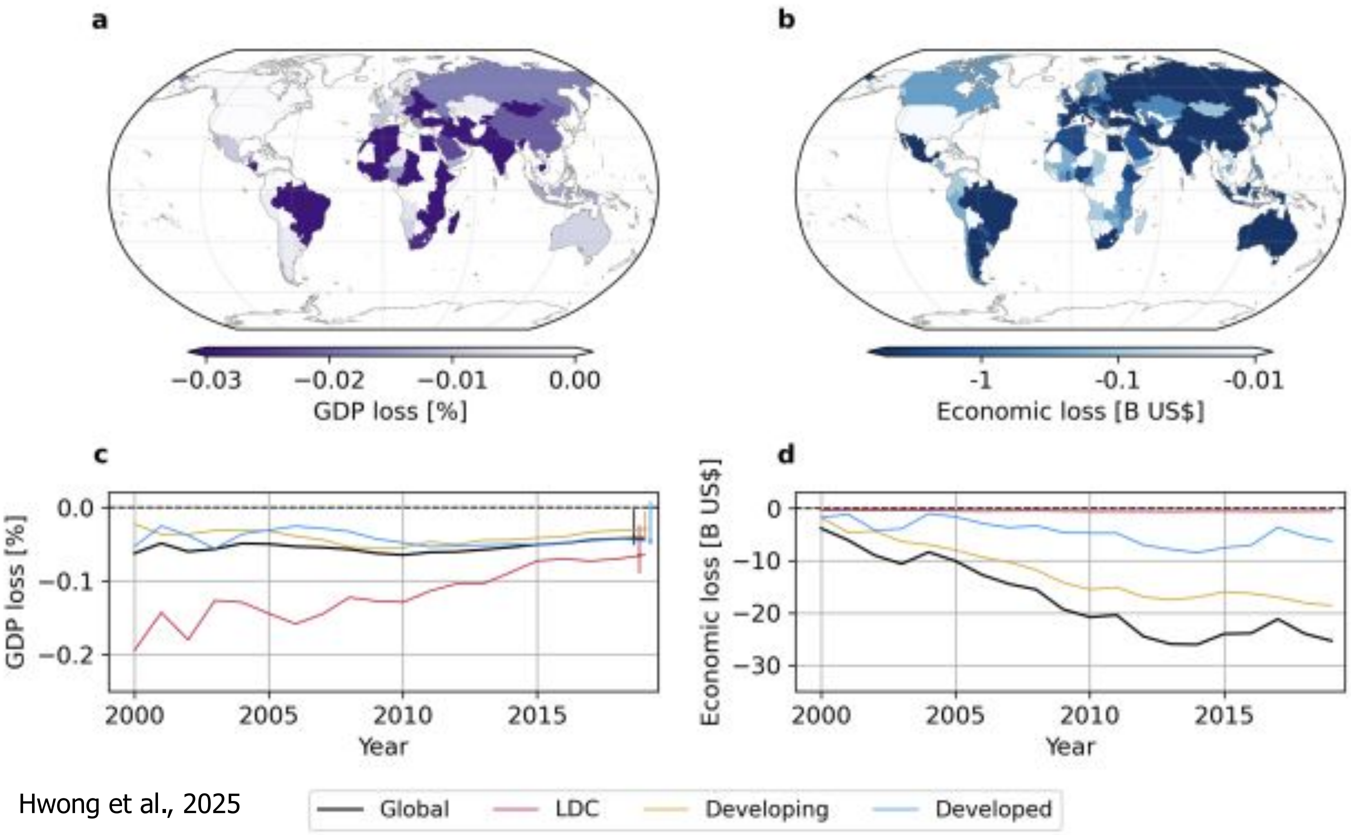
Loss rate: Modern agriculture can remove that same 1 cm in under 10 years.

Causes of degradation:

- Erosion by wind and water due to overgrazing
- Nutrient depletion from overtilling
- Compaction from heavy machinery
- Salinization from irrigation
- Urbanization and land sealing
- Chemical contamination



3.2 Disproportionate Economic Burden from Climate-Induced Agricultural Losses in Least-Developed Countries



Hwong et al., 2025

Global cost of land degradation: $878B/year (UNCCD, 2025)

Yields from degraded lands are 50% lower on average (MDPI, 2024)

Soil erosion costs U.S agriculture ~44B/year in lost productivity (WRI, 2020)

Market Failure and Externality

Negative Externality

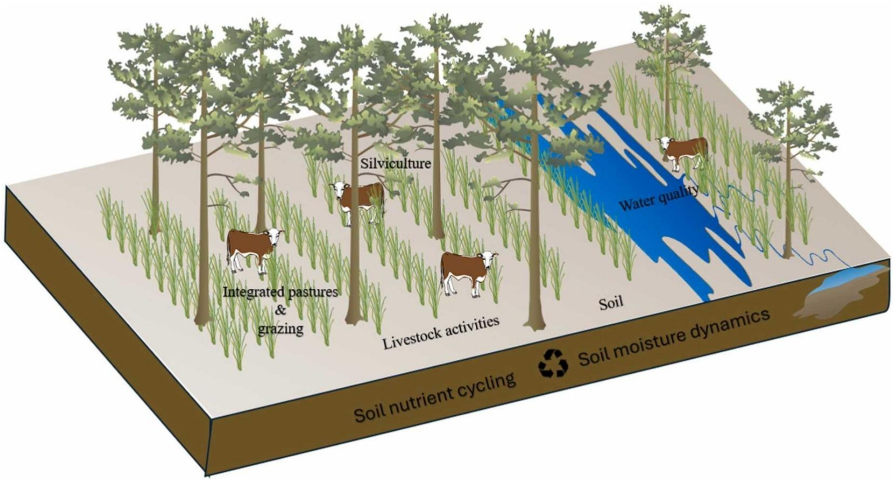
- Farmers internalize yield gains from intensive use
- Off-site erosion costs fall on water utilities, downstream farmers, and municipalities
- Future generations bear the full burden of depletion
- Time discounting makes near-term more attractive than soil conservation

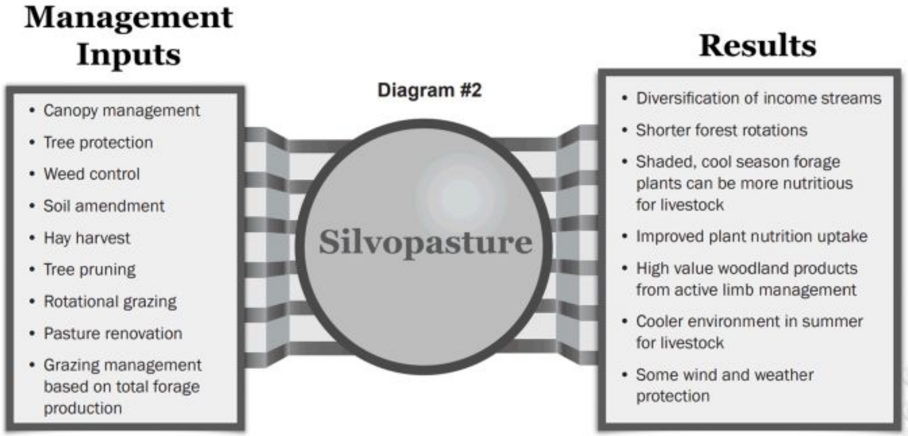
Tragedy of the Commons

- Healthy soil is a public good: carbon sequestration, flood mitigation, clean water
- These ecosystem services are not priced in markets
- Non-excludability means no private incentive to preserve

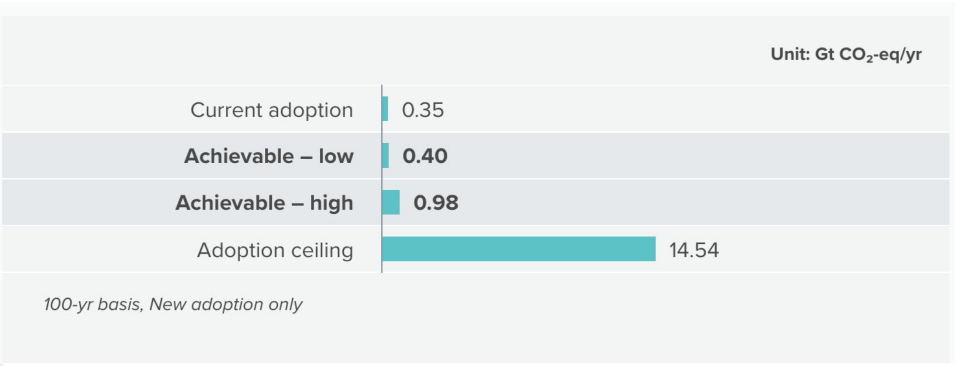
Silvopasture

The deliberate integration of trees, grazing livestock operations, and forage crops on the same land. (FS, 2025)





FS, 2025



Drawdown, 2025

- Cerrado lost ~50% of native vegetation (Pereira 2026); topsoil up to 40 tonnes/ha/yr (Carbon Pulse, 2025)
- Embrapa (government research agency) launched silvopasture trials across Mato Grosso & Goiás (Embrapa, 2024)
- Farmers integrate Brachiaria grass + intercropping maize plants (Souza, 2024)

Usual denominations	Degraded pasture Degraded savanna	Productive pasture	Degraded pasture	Degraded pasture	Degraded pasture
Proposed sustainable intensification / restoration	Natural or semi-natural pastures	Silvopastoral systems	Savanna restoration Silvopastoral systems	Savanna restoration Pasture restoration + Silvopastoral systems	Savanna restoration Pasture restoration + Silvopastoral systems
Proposed denominations	Cerrado	Productive pasture	Pasture with Cerrado Regeneration + Regenerating Cerrado	Biological degradation with Cerrado regeneration	Biological degradation + Severe Biological degradation
Vegetation profile					

Vieira, 2021

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